

Intelligent networks, standards and services

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The key principle of intelligent networks is the separation of the software that controls basic switch functionality, such as setting up switch paths, from the software that controls call progression to allow more advanced features to be rapidly introduced. This paper describes the current standardisation activities, the IN conceptual model and the IN components. It also includes a summary of some of the services available on an intelligent network, and some future developments.

1. Introduction

Over the last three decades, there have been a number of significant technological developments in public switched networks (PSTN), namely stored programme control switches, digital switching, ISDN and recently the introduction of intelligent networks (INs).

It takes several years to specify, develop, test and deploy new switch-based services, with additional features waiting for a time-slot in the switch development cycle. Also, the time taken to upgrade a large network with new software, and perhaps hardware, can exceed a year, constituting a considerable time lag from conception of a new service to the first customer using the service.

Hence, the major drivers for the addition of intelligence to the network are the:

- requirement for the rapid introduction of new services and features,
- reduction of the reliance on switch manufacturers for the provision of new services,
- ability to offer integrated service packages.

Through the deployment of intelligent networks telecommunications operators are endeavouring to meet the demands of these drivers.

The intelligent network development is essentially humanising the network — adding to the network the intimate knowledge of the user and those connected to it — taking the network ‘back’ to the days when Strowger thought the network too ‘intelligent’ and developed the automatic switch.

The following sections describe the current standardisation activities, the IN conceptual model and the IN components found in the network. Also included is an overview of some of the services implemented using an intelligent network. Some of the developments occurring in the field of intelligent networks are also outlined.

2. Intelligent network standards

A major step forward in intelligent networks was made when the Regional Bell Operating Companies (RBOCs) asked Bellcore to produce the advanced intelligent network (AIN) series of technical requirements. The development of AIN followed a series of multi-vendor discussions which took place during the late 1980s and which resulted in the production of AIN Release 0.0, leading on to the development of AIN Release 0.1 — with AIN Release 1.0 being the target AIN architecture for the next decade. Other standards bodies, for example the International Telecommunications Union — Telecommunications Standardisation Sector (ITU-T) and European Telecommunications Standards Institute (ETSI), have now also produced definitions for the intelligent network architecture using a range of conceptual and functional models, (see section 2.1). Figure 1 shows the relationships between the USA, European and international standards development process.

Figure 2 illustrates the ITU-T processes for the production of IN capability set 1 (CS-1), which forms part of the input to the definition of IN CS-2 and 3 along with contributions from ETSI in the form of ‘Core-INAP’ (intelligent network application protocol).

The IN CS-1 recommendations are in the Q.121x series of documents, with IN CS-2 contained in Q.122x. These

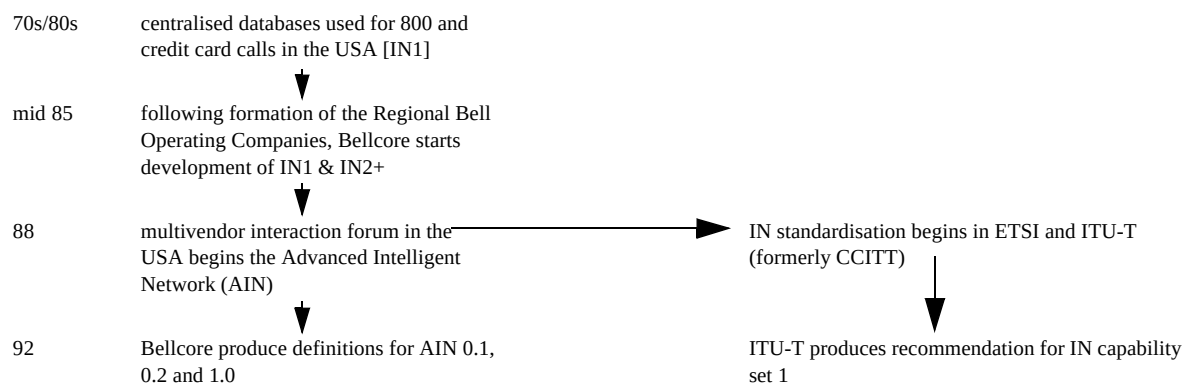


Fig 1 History of the development of AIN and IN capability set standards.

documents are developments from the base Q.1200 series of recommendations.

The Q.12xx recommendations provide the definition of the IN architecture through the use of a conceptual model.

2.1 The IN conceptual model

The IN concepts can be explained in terms of a model which provides a framework for the design and description of the IN architecture. This IN conceptual model (INCM) has been used as a tool during the development of the current IN capability sets. By explaining this model the IN concepts can be explained.

The IN conceptual model consists of four planes each of which provides a different abstract view of the capabilities of an IN structured network. This ranges from how services are defined, at the top layer, to how they are mapped to a physical network, the bottom layer (see Fig 3).

The INCM can be used to provide both a top-down view with respect to services and a bottom-up view with respect to networks.

The service plane

The service plane provides a service oriented view and describes services in terms of service features. The service features are themselves described in terms of global service logic. Each service feature is defined by a single set of global service logic (GSL). At this high level there is no view of the relationship of the service or service feature with the physical network on which it is deployed.

In a high-level service creation environment something representing the service feature level of granularity for creating new services would be expected. How fine or coarse this granularity is has not yet been the subject of standardisation work. The service plane was not standardised for IN CS-1, i.e. no Q.1212 was produced.

The global functional plane

The global functional plane models an IN structured network as a single entity. It describes globally available service-independent building blocks (SIBs) (see Table 1).

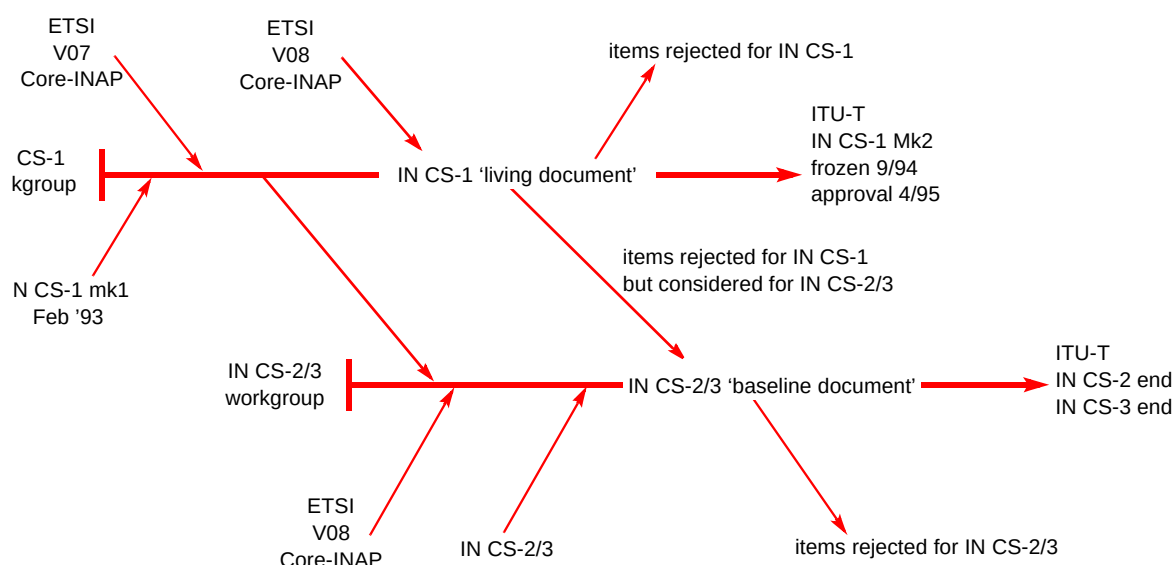


Fig 2 ITU-T process for the production of IN CS-2 and 3.

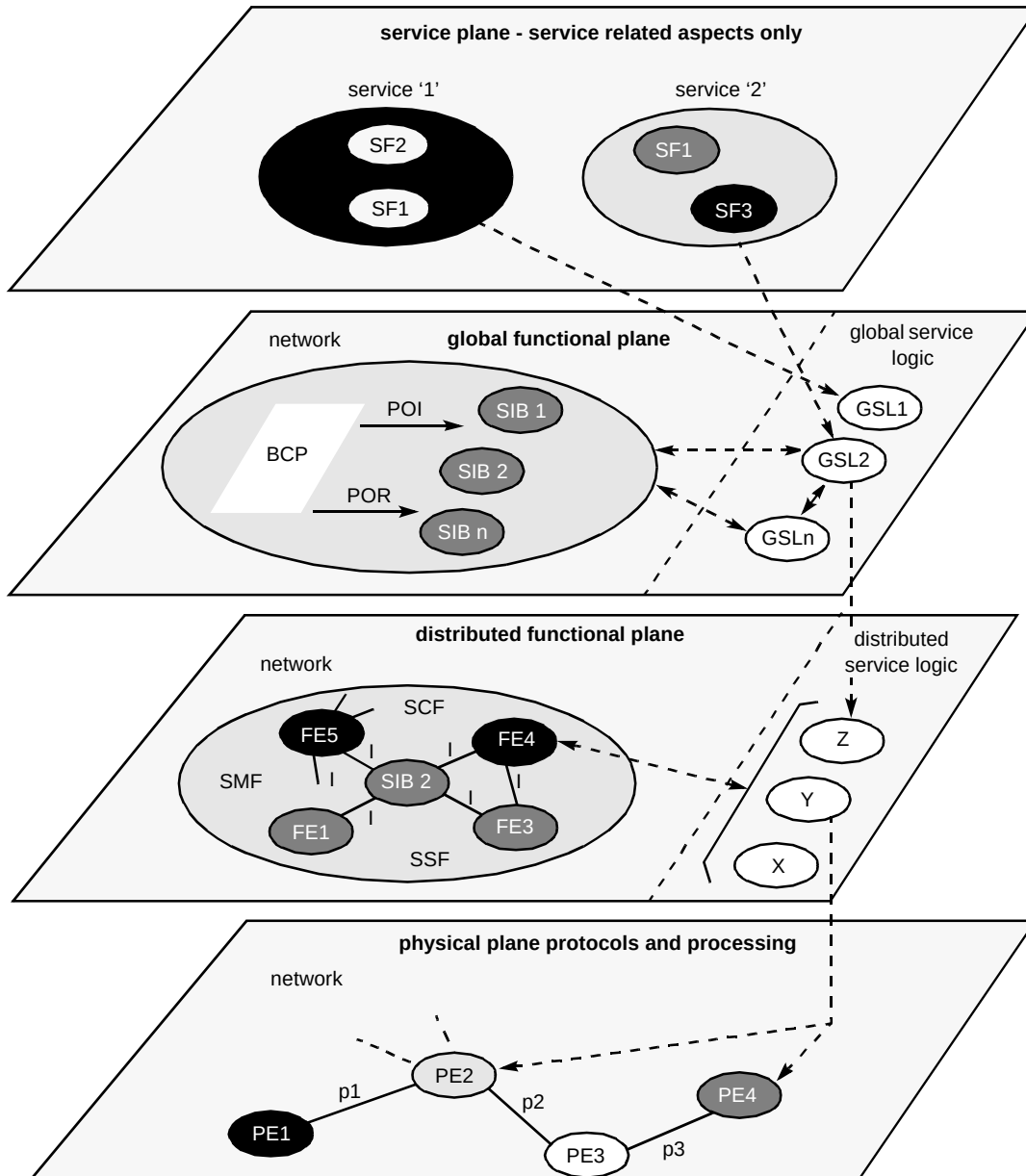


Fig 3 The IN conceptual model (based on Q.1201 [1]).

These SIBs are representations of the basic capabilities of the underlying functional network. There is one special SIB, the basic call process (BCP), with which the other SIBs interact. Also defined are the points of interaction between the BCP SIB and the other SIBs. A set of global service logic defines in terms of SIBs a single service feature. The GSL describes where and how SIBs interact with the BCP.

The distributed functional plane

The distributed functional plane models a distributed view of an IN structured network. It is the first point where a traditional network architecture can be seen with the network being defined in terms of functional entities.

Table 1 Service-independent building blocks (Q.1213).

Algorithm	Limit	Service data management	Verify
Charge	Log call information	Status notification	Basic call process (BCP)
Compare	Queue	Translate	
Distribution	Screen	User interaction	

Each of the functional entities is able to perform actions. These are described as functional entity actions (FEAs). An FEA is the atomic unit of action and is wholly contained within a functional entity. FEAs are themselves comprised of functions (elementary functions (EFs)) which are the atomic unit for FEAs.

Service-independent building blocks are realised within this plane as a sequence of FEAs. Each SIB is described by a set of distributed service logic which describes this sequence of FEAs. Ultimately some SIBs require information to be passed between functional entities to allow an SIB to be realised. It is these information flows that give rise to the protocol (INAP) in the physical plane (see Fig 4).

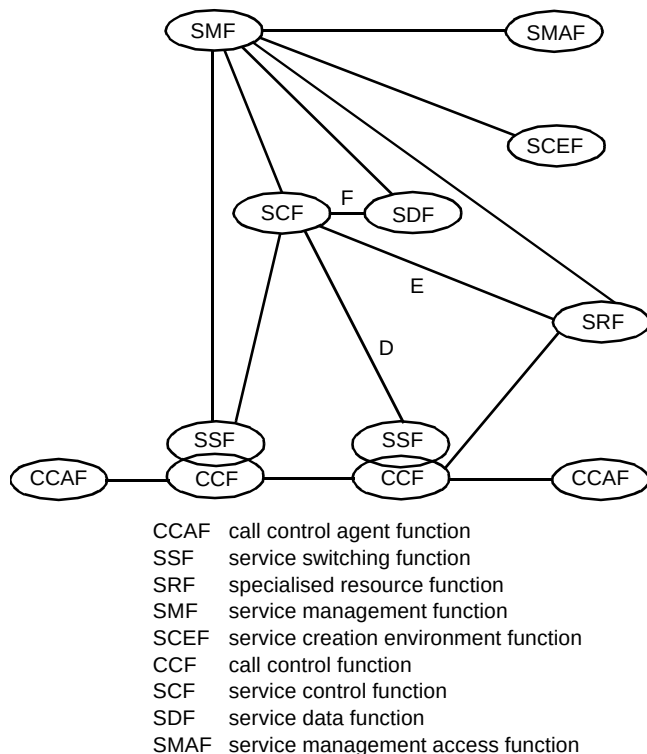


Fig 4 ITU-T defined functional IN architecture (distributed functional plane, based on Q.1211 [1]).

The physical plane

The physical plane resolves the functional plane into a physical network and models the different physical entities and protocols that exist within an IN structured network.

For all the planes in the IN conceptual model there is a one to many relationship as you descend, i.e. a service is comprised of one or more service features; each service feature contains one or more SIBs as illustrated in Fig 5.

Mapping the elements of the INCM

Figure 5 shows how the various elements of the INCM map.

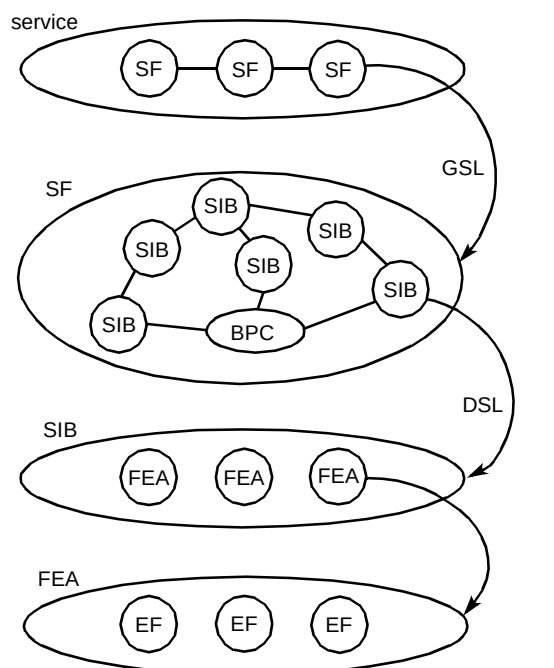


Fig 5 Mapping of elements in the IN conceptual model.

2.2 The IN functional architecture

The distributed functional plane can be re-drawn to illustrate the IN functional architecture (Fig 4), where the FEAs are grouped in to functional areas, such as the service switching function, service control function, etc, which communicate through the use of information flows across reference points (reference points D, E and F are shown in Fig 4).

ETSI, in 1993, produced ‘Core-INAP’ which is an operator implementable version of the intelligent network application protocol for use in the D, E and F interfaces. The protocol is message-based and is defined in abstract syntax notation 1 (ASN.1) [2]. The other interfaces shown in Fig 4 have not been standardised in IN CS-1.

3. Intelligent network components

Intelligent network functional entities, such as the service switching function, can be partitioned into a number of different physical elements, as shown in Table 2. This does not imply that other combinations of functional entities into a physical entity not shown should be disallowed.

The following sections provide descriptions of some of the elements identified in the table, such as the service switching point, service control point and service node.

Table 2 Mapping of functional entities to physical entities (Ref: Q.1205).

	SSF/CCF	SCF	SDF	SRF	SMF	SCEF	SMAF
SSP	Mandatory	Optional	Optional	Optional			
SCP		Mandatory	Optional				
SPD			Mandatory				
IP	Optional			Mandatory			
Adjunct		Mandatory	Mandatory				
Service Node	Mandatory	Mandatory	Mandatory	Mandatory			
SSCP	Mandatory	Mandatory	Mandatory	Optional			
SMP					Mandatory	Optional	Optional
SCEP						Mandatory	
SMAP							Mandatory

Figure 6 illustrates one possible IN configuration where each functional entity is mapped directly to a physical entity.

3.1 Service switching point

The service switching point (SSP) comprises the call control function (CCF) and the service switching function (SSF), as shown in Fig 7. The CCF provides the call processing for basic telephony switching functions. The SSF,

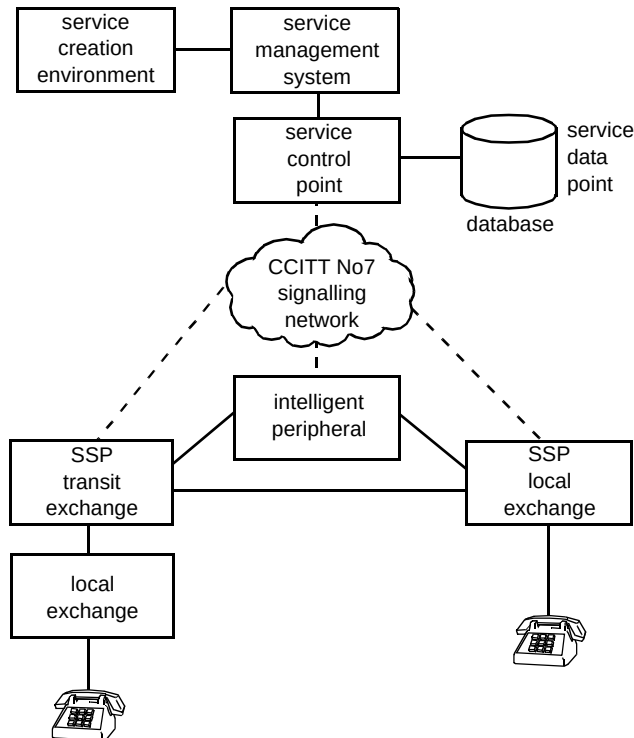


Fig 6 An intelligent network architecture.

which is associated with the CCF, provides the interface to other physical entities, such as the service control point (SCP) and intelligent peripherals (IPs).

The SSP, as shown in Table 2, can optionally contain a service control function (SCF), specialised resource function (SRF) or service data function (SDF). If the SSP is the local exchange it will also contain the call control agent function (CCAF).

The separation of enhanced call control run on the SCP from service switching allows the progression of the call to be suspended while further information regarding how the call is to be handled can be obtained from the service data point (SDP). The decision to suspend call processing is based on the meeting of pre-specified criteria, i.e. triggers,

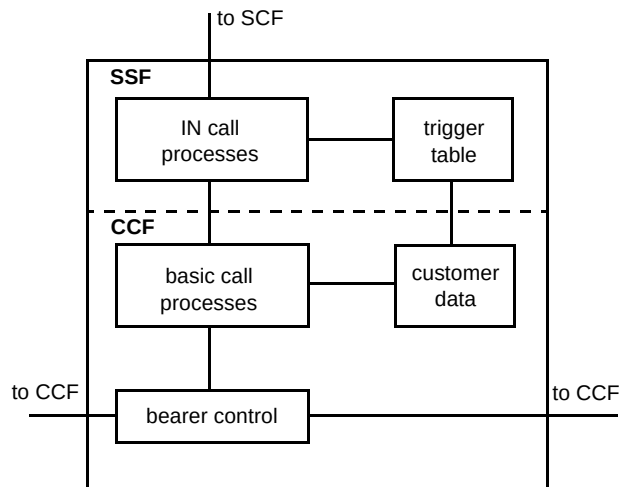


Fig 7 Service switching point (SSP).

for example dialled digits or line condition, at certain points during the call.

As shown in Fig 6, SSP functionality can be provided at both local and trunk exchanges, the main differences being the different triggers available at each exchange.

IN call model

In the separation of the call-control processing from the service switching functions in the SSP, the control system needs to have a reference model for the call from which enhanced service logic can be instigated, and also returned to, following the invocation of service logic. This model is referred to as the IN call model.

The IN call model is based on the definition of a basic call state model (BCSM), which defines the progression of the call in terms of points in call (PICs) and detection points (DPs). BCSMs are finite state models which reflect the originating and terminating sides of the call referred to as O_BCSM and T_BCSM respectively. The BCSMs as

defined by ITU-T CS-1 are shown in Fig 8, with the convention for illustrating BCSMs shown in Fig 9. Call models also exist for the Bellcore-defined AIN. The differences between these call models add to the difficulty of interworking intelligent network equipment from different vendors

3.2 Service control point

The service control point (SCP) is a general-purpose computing platform that hosts the advanced service control software or service logic.

The SCP should provide high throughput, high reliability/availability, fast response times and access to a high capacity database system. The SCP is usually a high throughput real time computing platform on to which is loaded the SCP node software and a service logic execution environment (SLEE) (see Fig 10) [3]. The SLEE shields the service logic from the underlying node/platform operations. The SCP node software provides the common utilities such

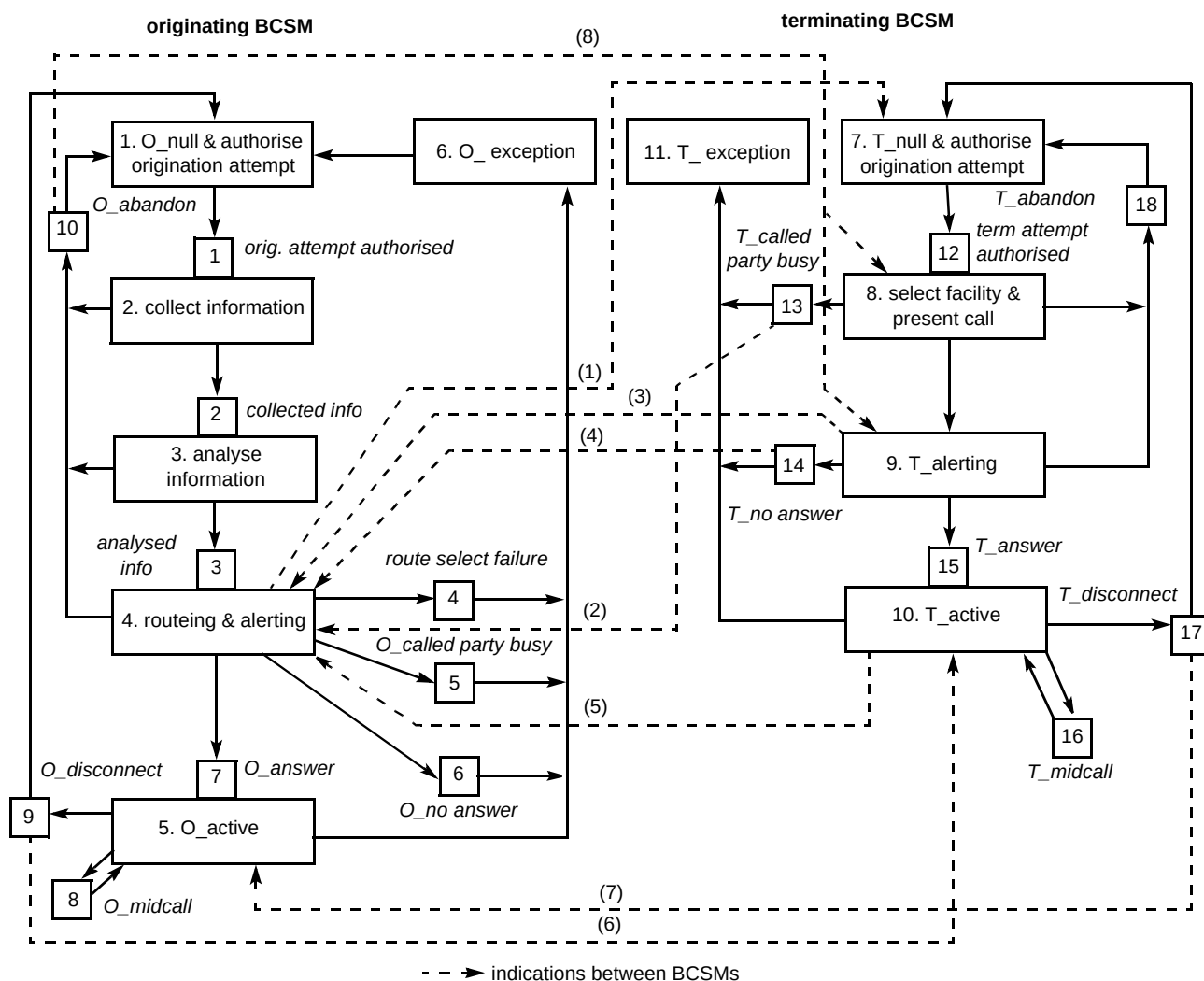


Fig 8 ITU-T basic call state models for originating and terminating half calls (based on Q.1214 [1]).

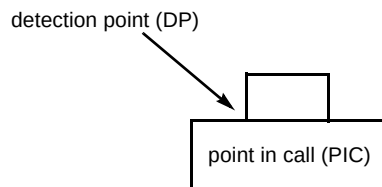


Fig 9 Point in call and detection point illustration.

across a number of SSPs in an intelligent network architecture. Under the control of service logic a caller would interact with the service node using each feature or service as required. The service node contains all the necessary logic and resources to deliver services to the customer.

The service node provides an alternative service platform to the centralised SCP with network based IPs.

The service node can be used as an 'edge-of-network' solution for services such as voice-mail. The system can also be used for number translation services where the service node originates the new call leaving the original switch in the speech path.

The BT service node is described in detail by Kabay and Sage [4].

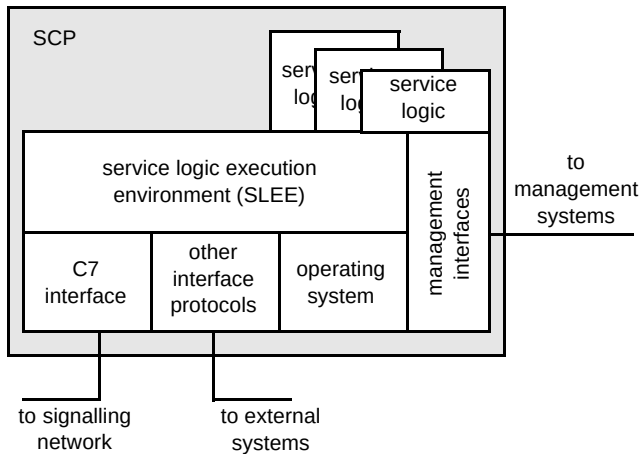


Fig 10 Service control point.

as signalling system support, data-base communications, transaction monitoring and alarm reporting.

Service logic is derived from the manipulation of a range of software building blocks (see section 2.1) within the service creation environment (SCE). It is the responsibility of the management system to deploy the resultant service logic to the network.

An SCP can be positioned in several locations in the network, for example connected to a number of SSPs via the signalling network, as in Fig 6, or directly connected to the SSP as an adjunct processor via a high-speed interface.

3.3 Service node

The service node can provide intelligent network services, and engage in flexible information interactions with users. It communicates directly with one or more service switching points (SSP), each with a point-to-point signalling and voice circuit connection. Functionally, the service node contains a service control function (SCF), service switching point (SSF), service data function (SDF) and specialised resource function (SRF).

It provides a high availability, multipurpose platform of fully integrated resources (see Fig 11), such as voice announcers, speech and DTMF recognisers, network-based answering, etc, that can be utilised during a call and shared

3.4 Intelligent peripheral

Intelligent peripherals (IPs) provide any additional specialised resource functions (SRFs) required in the network for the operation of services (see Fig 12). The IP can also contain a service switching function/call control function (SSF/CCF) to provide switching to the appropriate resource in the IP.

Examples of IP resources include:

- customised and concatenated voice announcements,
- DTMF digit collection,
- synthesised voice/speech recognition,
- audio conference bridge,
- information distribution bridge,
- tone generator,
- protocol converters ,
- text-to-speech synthesis.

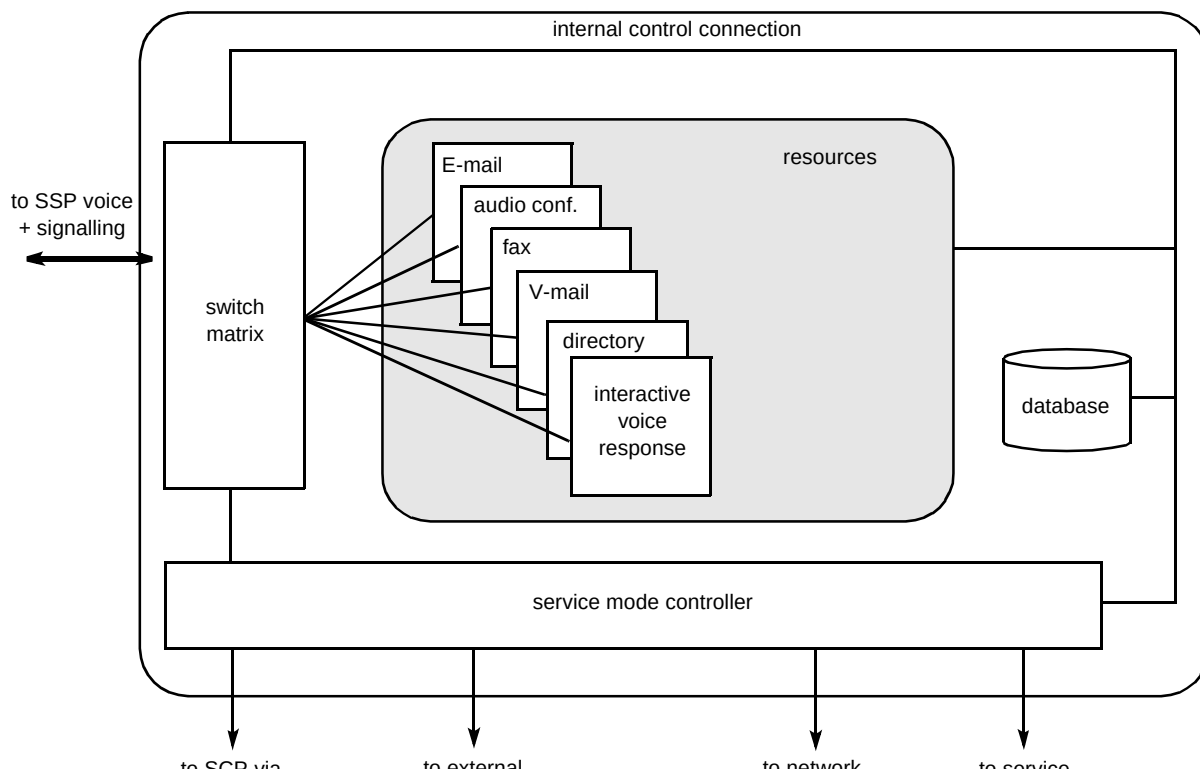


Fig 11 Service node.

Depending on the actual network architecture, the IP can be either directly controlled by the SCP via the signalling network, with voice circuits connecting the IP to the SSP, or controlled from the SCP via the SSP.

The IP can be connected to more than one service switching point, allowing sharing of IP resources in the

network — particularly important for expensive/specialised resources. The actual location of the specialised resources in the network depends on a number of factors, including cost, service operation, and performance.

3.5 Signalling network

The signalling network provides the communications between the component parts of the intelligent network, and is used to pass the controlling messages between the SCP, SSP, SDP (service data point) and IP. The IN application protocol (INAP) provides a standard message set for use in IN services, and is used for the interfaces between the SCP, SSP, IP and SDP. The protocol makes use of the non-circuit-related transaction capability provided by the transaction capabilities (TC) and signalling connection control part (SCCP) of the C7 signalling system (see Fig 13). INAP is described in Abstract Syntax Notation No. 1 (ASN.1) in ITU-T Q.1218 and ETSI 'core-INAP' [2, 5, 6].

3.6 Service creation environment

The ITU-T definition of IN CS-1 does not include standards for the service creation environment function (SCEF) in the same level of detail as for the service control point for example, but does identify the function and the links to the other functions, e.g. the service management function (SMF).

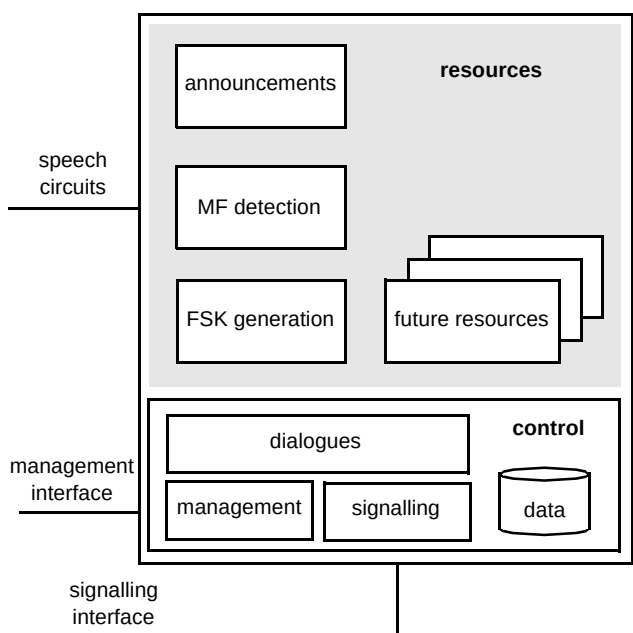


Fig 12 Intelligent peripheral.

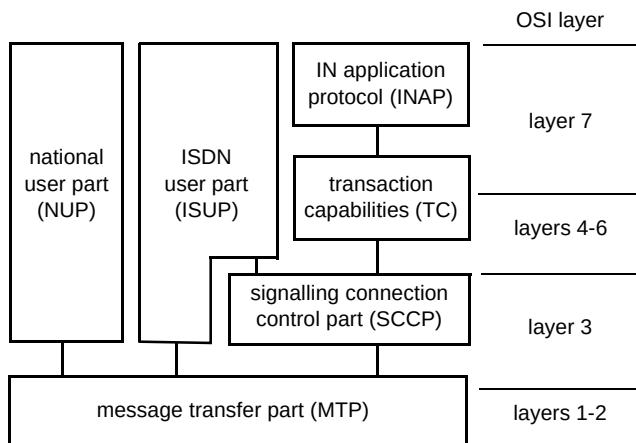


Fig 13 CCITT signalling system No. 7.

The service creation environment (SCE) provides tools for the rapid creation of new services using object oriented programming techniques and reusable building blocks through the use of a graphical user interface (GUI). Also provided is a means to simulate the service before downloading to the network.

Through the use of SCE tools, the time taken to design a new service can be greatly reduced from that taken to implement new switch-based services, but in order to make this achievable the SCE tools must use building blocks which have already gone through rigorous testing.

Currently, service creation environments are closely coupled to the service control point with the SCP vendor providing the SCE which is matched to the architecture of their SCP.

A description of the processes and tools used in the field of service creation can be found in Turner [7].

3.7 Service and network management systems

As with the service creation environment function, the service management functions are not standardised in ITU-T IN CS-1.

Currently, the standards bodies are investigating the harmony between functions in the intelligent network architecture and those defined in the telecommunications management network (TMN) standards.

The service management system, which is a combination of the service management function (SMF) and the service management agent function (SMAF), provides an operations support system, responsible for service management, deployment, provisioning of customers and updating customer-specific data held on the SCP and service database. Additional functions which could be performed are database administration, network surveillance and

testing, network traffic management and network data collection.

A description of service and network management functions related to intelligent networks can be found in Holmes and Street [8].

4. IN supported services

In the development of the IN CS-1 conceptual model, Q.1211, a number of example, or benchmark, services were used to identify the capabilities to be provided by IN CS-1. These services are often referred to as the IN CS-1 services, though they themselves are not standardised. They are only used for the validation of the concepts and should not be viewed as a limit to the services supported through an IN CS-1 architecture (see Tables 3 and 4).

Table 2 IN CS-1 benchmark services.

Abbreviated dialling	Destination call routing	Selective call forward on busy/do not answer
Account card calling	Follow-me-diversion	Split charging
Automatic alternative billing	Freephone	Televoting
Call distribution	Malicious call identification	Terminal call screening
Call forwarding	Mass calling	Universal access number
Call re-routing distribution	Originating call screening	Universal personal telecommunications
Completion of call to busy subscriber	Premium rate	User-defined routing
Conference calling	Security screening	Virtual private network
Credit card calling		

Table 3 IN CS-1 service features.

Abbreviated dialling	Call queuing	Customised ringing
Attendant	Call transfer	One number
Authentication	Call waiting	Origin dependent routing
Authorisation code	Closed user group	Originating call screening
Automatic call back	Consultation calling	Originating user prompter
Call distribution	Customer profile management	Personal numbering
Call forwarding	Customised recorded announcement	

The IN CS-1 benchmark services are in the category 'single ended', 'single point of control', and are referred to as Type A services. Currently, the ITU-T are developing standards for Type B services, which will form part of IN CS-2.

There is a view that there are no real IN services, just services which are better supported in an IN manner. This can be as a result of the nature of the service or management of service data.

IN supported services have one or more of the following characteristics:

- networked services — services that require co-ordinated action across a number of nodes,
- services that require large databases and real time updates (especially by the customer),
- services that require tailoring to meet individual customer requirements,
- speculative services where the market demand is unclear and for which a trial platform is required.

One of the advantages of offering services in an IN manner is the reduction of the interaction between service features, which could cause unpredictable or confused operation. This is facilitated through the implementation of services on a common platform. Also, as the services are offered using a common platform feature, the interworking can be improved, leading to combinations of services or packages operating together, which reduce the need for multiple implementations of the same service for each platform in the network.

Some example IN services (described in the following sections) are:

- number translation,
- virtual private networks,
- cashless services,
- wide area centrex,
- automatic call distribution,
- personal numbering,
- call completion services.

4.1 *Number translation services (freefone and pre-mium rate)*

One example of a number translation service is the provision of a single, fixed number for customers with geographically dispersed locations. Incoming calls will then be routed according to a predetermined schedule, taking into account, for example, nearest location, time of day or day of week.

Figure 14 illustrates an intelligent network implementation of a simpler number translation service, where the SCP performs a database look-up to determine the appropriate destination number. The translated number is then sent to the SSP for the connection to the destination.

Number translation services are often linked to variable charging services, such as BT's Free-Call and Lo-Call, and CallStream services.

4.2 *Virtual private networks*

Virtual private networks (VPNs) allow users to create networks which can be considered private in terms of the flexibility, re-configuration, security, availability and exclusivity without having to manage dedicated private lines.

VPNs can be defined as a logical closed user group implemented on the public network, with the following main characteristics [9]:

- provision of a private numbering plan,
- call charging on the basis of duration usage as opposed to rental for private circuits,
- usage for switched voice, switched data or both.

Advantages of VPNs include a rich feature set (for example, call screening, authorisation codes, flexible routing), simplified network management tasks for users due to outsourcing, advanced billing facilities and a single interface for domestic and international requirements.

VPNs can be accessed either directly through a dedicated link (on-net) or through a switched connection from the PSTN (off-net). Off-net access usually requires the use of special access numbers and authorisation codes.

Virtual private network services include switched voice, switched data, private line, travellers or calling card access and 0800 access.

4.3 *Cashless services*

Cashless services, also known as alternative billing services (ABS) or calling card in the US, allow the customer to place calls which are then charged to different accounts. An example of this is the BT Chargecard service.

4.4 *Wide area centrex*

The term centrex is used to describe the provision of modern PABX services over the PSTN. Wide area centrex

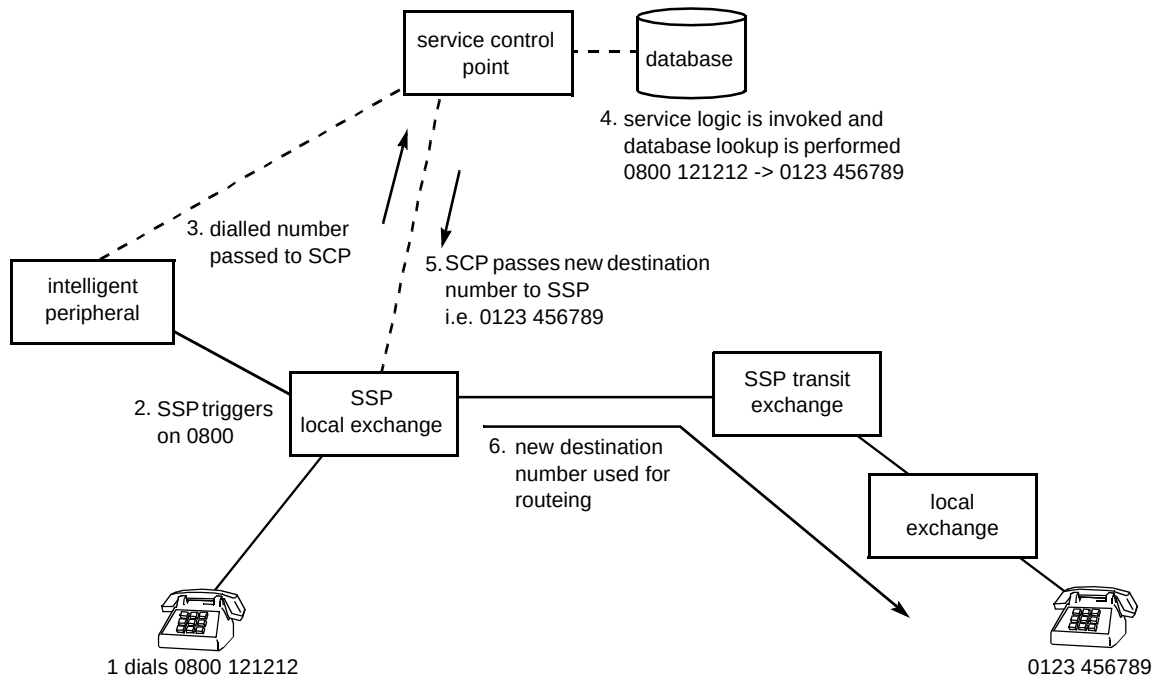


Fig 14 Number translation service illustration.

refers to the scenario where the customer has sites covered by two or more network nodes. A unified numbering plan and transparent operation of services are key features of wide area centrex.

4.5 Automatic call distribution

Automatic call distribution provides the distribution of incoming calls to a number of separate answering points, these may also be geographically separate, either on a fixed proportional basis or dependent on availability of answering points. This service is widely used by ticket booking agencies, operator positions and airlines, and is often associated with a single directory number.

4.6 Personal numbering

This refers to the provision of a single number for the customer, with the network providing the capability of call completion based on an updatable personal profile held in the network. The call could be completed to the person's actual or most likely location with a voice mailbox or paging system being the default option if the person is busy.

4.7 Call completion services

These are simple diversion services, such as divert-on-busy or divert-on-no-reply, and those already offered as switch-based services, for instance BT's network services.

5. Example of IN implementation

Many telecommunications operators have installed or are planning to install intelligent network equipment to meet the needs for specific service requirements, such as freephone, premium rate services and calling card services.

An example of an early IN implementation is the BT digital derived services network (DDSN) which provides the advanced linkline and callstream services based on AT&T IN technology and installed in 1989. The DDSN consists of an overlay network of 13 AT&T 5ESS switches acting as SSPs and three pairs of network control points (NCPs) providing the service control function.

The advanced linkline service includes the following features:

- time-of-day/day-of-week routing,
- geographical routing — routing depending on area where call originates,
- proportional routing — based on a predetermined routing plan,
- call queuing/completion of busy calls,
- call prompting — uses voice guidance and digit collection.

In 1992, IN Phase 1, an SCP consisting of a GPT System X front end and a Tandem computer, was

introduced into the BT network. The IN Phase 1 platform is used to provide basic number translation services.

6. The future intelligent network?

In the standards arena, the harmonising of the developments of AIN, ETSI and ITU-T standards will begin to occur. This may ease the possible problems when interworking equipment in a multivendor environment.

Significant developments are occurring in the field of speech recognition, which, linked with IN capabilities, will move services offered on the network towards the goal of a network to which you can talk. Speech recognition could be initially used in the provision of a voice dialling system, leading on to more interactive-based services, making use of word-spotting capabilities of the systems.

Looking to the future, it should be noted that some people do not consider the standards-based IN described here as the intelligent network in the sense that it has not yet re-created the user-friendly interface of the operator-based networks with a network to which you talk and which talks back, offering you what you want when you want it [10].

Intelligence can exist in a number of locations in the network; outlined so far has been the use of a centralised IN architecture, but this excludes the intelligence found in modern customer premises equipment (CPE), such as smartphones, PBXs and computer telephony integration (CTI), etc. As intelligence in the network becomes more complex, intelligent CPE will be required to make use of some of the enhanced features. A key issue with the distribution of intelligence is that of feature interworking and interaction (desirable and undesirable respectively) in a distributed processing environment.

IN concepts, though currently applied to telephony-based networks, are also applicable to other traffic types, such as data, broadband, video and multimedia services. Over the next few years this area will see significant attention to development and growth.

With the transformation of the telephony network to a 'data' network and the increase in information networking, the form of future intelligent networks will be different from those of today, implemented through a process of both evolution from the existing installed platforms, and revolution via the integration of new technologies and concepts.

Acknowledgement

Acknowledgement is made to the ITU-T, as the copyright holder, for the reproduction of Figs 3, 4, and 8 [1].

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Alan Munday, originally a chemistry graduate, has spent the whole of his professional career working on computer and data network systems.

He joined BT in 1991 and has spent his time working within what is now the network intelligence design unit.

He has specialised on IN CS projects and is currently working on IN evolution requirements for the narrowband network and IN requirements for the broadband network.